

CityGaze: A Human-AI Gaze-Informed Workflow for Context-Aware Social Sharing on OHMDs

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System Overview

CityGaze enables **content creation** and **context-aware discovery** on OHMDs via gaze, speech, and lightweight confirmation.

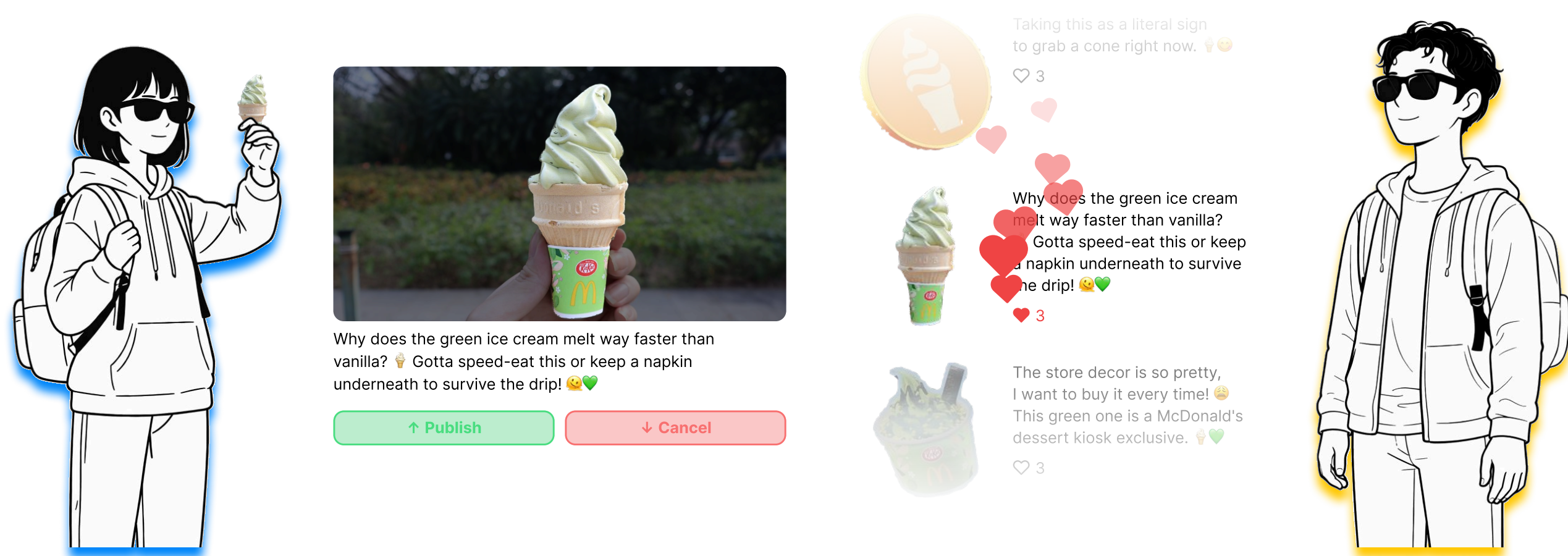


Fig. 1: Left: Alice creates a *Glimpse* via gaze + speech. Right: Bob discovers it in a heads-up feed.

Motivation & Problem

OHMDs still lack interfaces for **situated social sharing**, and existing solutions transplant phone-like paradigms.

Goal: Spontaneous and seamless sharing and browsing of social media content on OHMDs with limited interruption on the go.

Glimpse Creation: “Look-and-Speak”

A compact **heads-up** authoring pipeline:

1. **Capture** – FPV frame + gaze coordinates.
2. **Segment** – SAM generates multiple object masks.
3. **Refine** – speech + LLM (Gemini) finalize mask and caption.
4. **Publish** – ring confirmation, LLM adds category and tags.

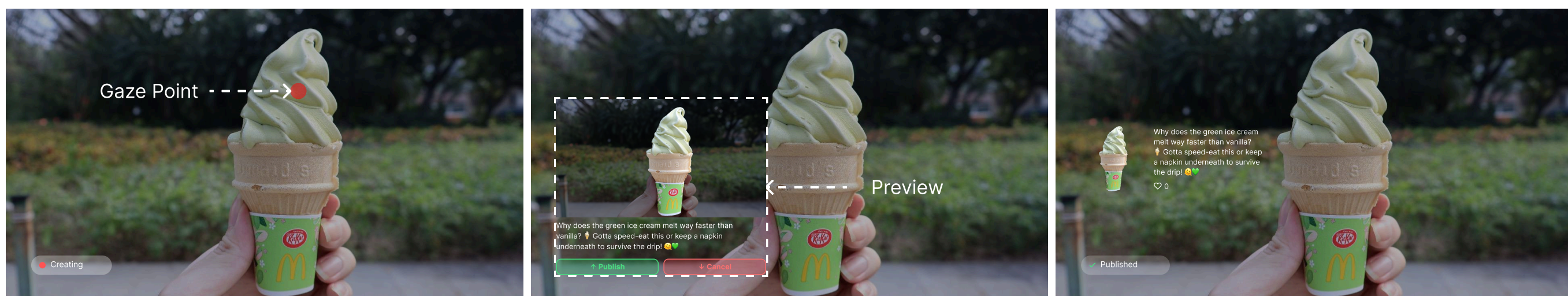


Fig. 2: Creation workflow.

Glimpse Consumption: Context-Aware Feed

Two-Stage Filtering Pipeline

Stage 1 – Visual Context Change Detection

- CNN (MobileNetV3-Small) encodes frames at ~ 3.3 Hz.
- 16-frame window, scene change flagged when $\geq 80\%$ of frames show cosine similarity < 0.6 .

Stage 2 – Distance + Semantic Filtering

- Haversine distance ≤ 30 m.
- LLM (Gemini) matches tags to current context.



Fig. 3: Peripheral feed with auto-scroll.

Contributions

- 1 **CityGaze** system for context-aware OHMD social sharing.
- 2 **Look-and-Speak** human-AI co-creation pipeline.
- 3 **Two-stage delivery** via geo + visual-semantic filtering.
- 4 **Empirical findings** from two user studies.

Formative Study

Part 1: Creator-Side (N=5)

100 paired observations – co-creation better preserved **intent** than AI-only ($W = 213.5$, $p < .001$)

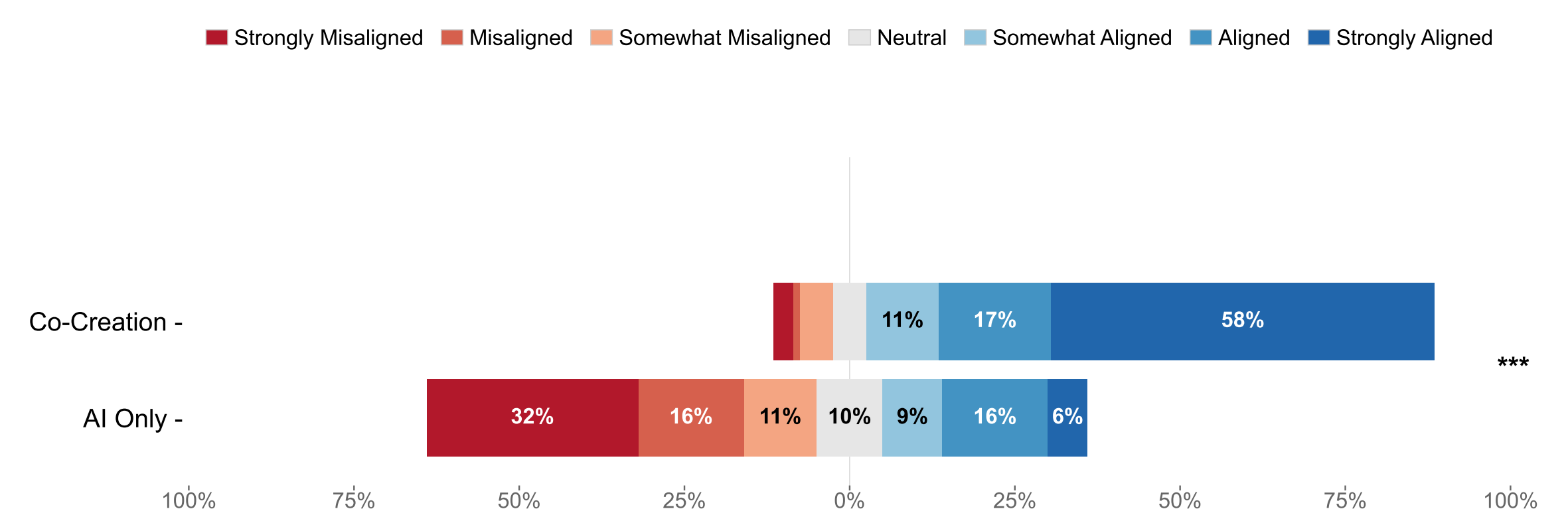


Fig. 4: Intent alignment ratings.

Part 2: Viewer-Side Ablation (2 x 2, N=12)

4800 instances. **Highlight** is the primary driver of perceived quality.

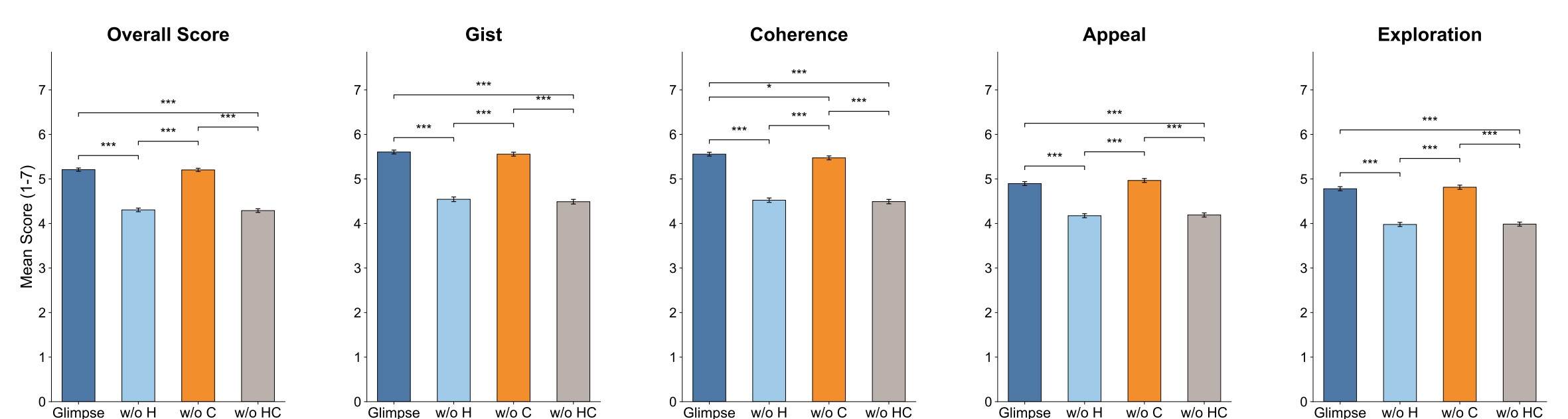


Fig. 5: Ablation results across four conditions.

System Architecture

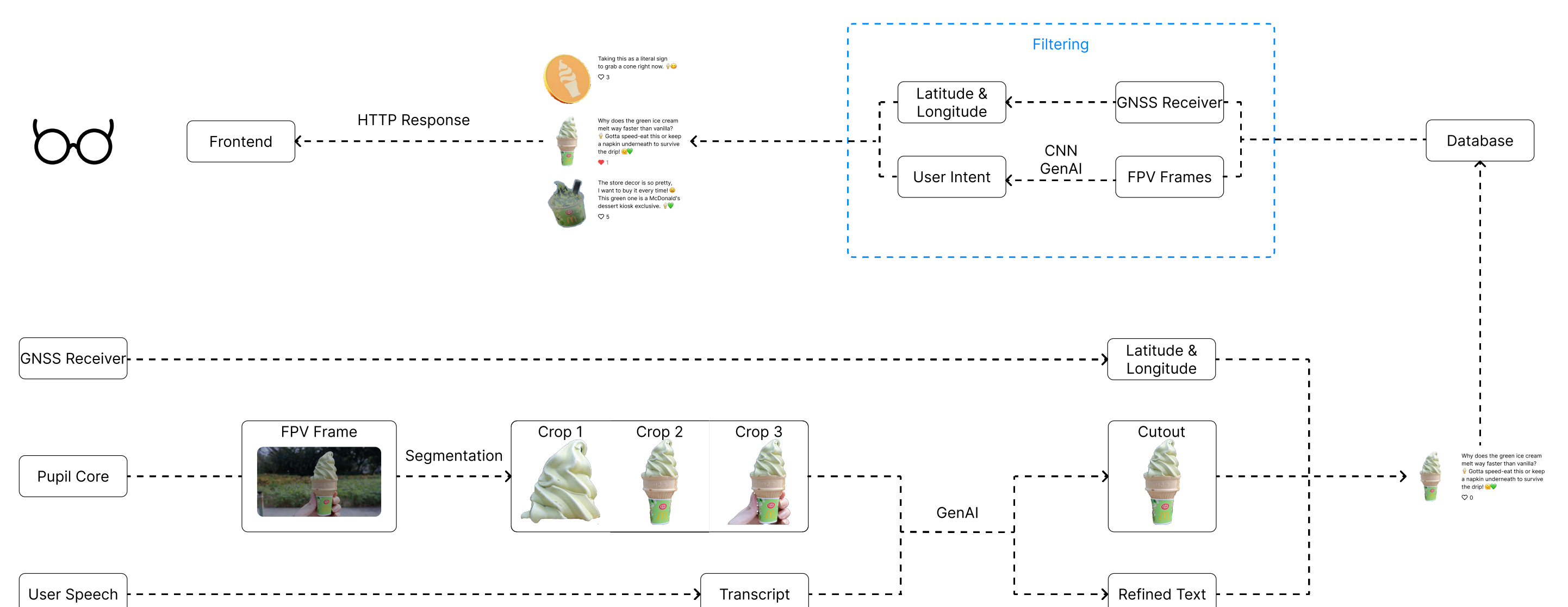


Fig. 6: Client-server architecture. Sensor streams (GNSS, FPV frames, voice) are processed by the back end, and results are rendered on the OHMD.

Exploratory Study

Study Design

N=12, ~ 90 min/session, 800 m route across **4 urban scenes**. 36 pre-seeded *Glimpses*, each participant created ≥ 3 new ones.

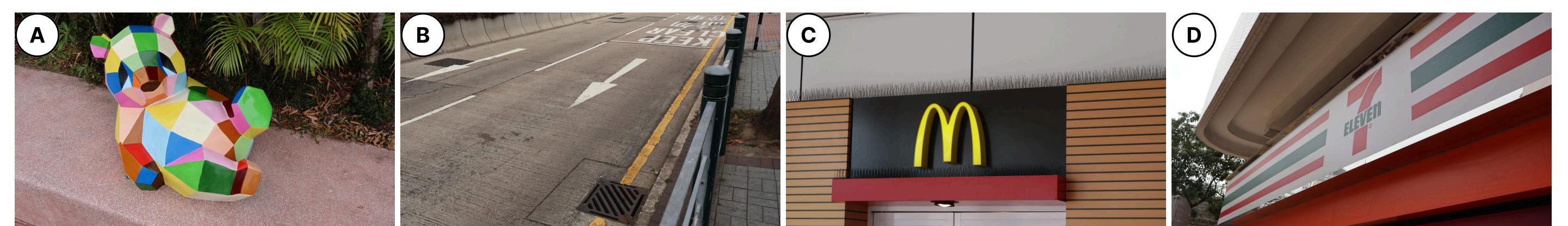
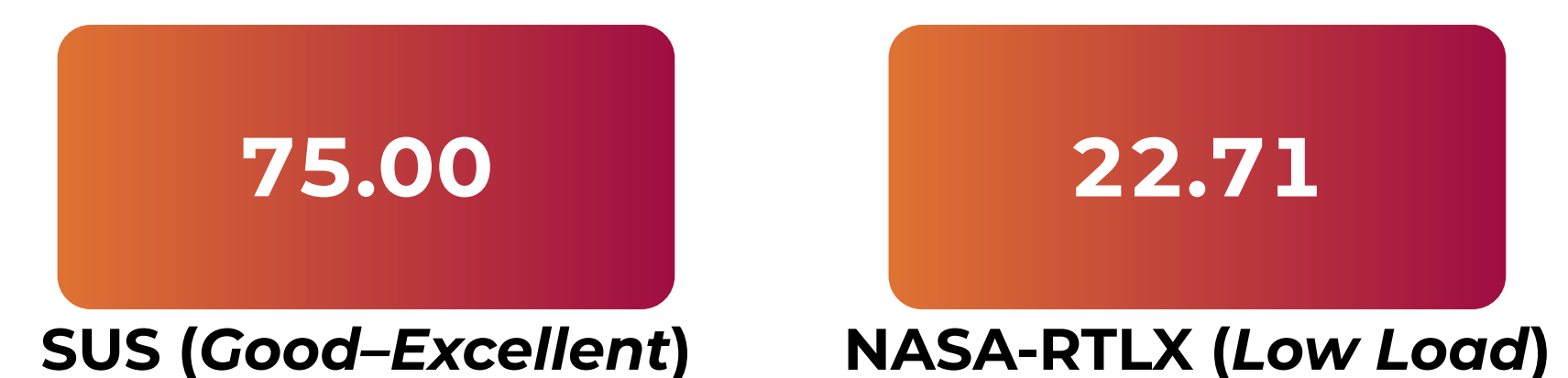


Fig. 7: 800 m urban study route across 4 scenes.

Quantitative Metrics



Key Findings

RQ1 – Engagement

- Brief speech became rich, grounded posts (spontaneous creation).
- Digital prompts engagement with the physical world.

RQ2 – Socio-Spatial

- Digital co-presence mitigated solo-travel loneliness.
- Balanced attention between digital content and surroundings.